

Sr Data Engineer

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Professional Summary:

- ▶ Over 11+ years of overall IT experience, diversified exposure in Software Process Engineering, developing, building enterprise applications using Python, PySpark, and Big Data Technologies in various domains like Financial, Retail and Healthcare sector.
- ▶ Extensive experience on different Big Data Platforms like Cloudera (CDH), Azure, AWS and GCP. Strong experience in all aspects of big data management, including ingestion, storage, querying, processing, and analysis.
- ▶ Leveraged Python, Spark Core, Spark SQL, Data Frames to develop high-performance data processing applications, achieving significant data processing speedups.
- ▶ To fulfill business intelligence requirements, scalable LookML models—including views, explorations, dimensions, and measures—were designed and put into use.
- ▶ ELT pipelines in GCP BigQuery were designed and tuned to increase the scalability and efficiency of data processing.
- ▶ Used Microsoft SQL Server and Power BI to design, implement, and maintain sophisticated BI solutions that allowed for data-driven decision-making. successfully converted intricate business requirements into LookML designs that matched the reporting and visualization expectations of stakeholders.
- ▶ proficient at creating and implementing scalable, cloud-based machine learning solutions using GCP services including BigQuery, Vertex AI Platform, Cloud Storage, AutoMLOps, and Dataflow.
- ▶ ELT pipelines in GCP BigQuery were developed and maintained to facilitate smooth analytics, data transformation, and integration.
- ▶ Expertise in Azure Cosmos DB, ensuring high availability and performance for ML-driven applications.
- ▶ Hands on experience working on different Hadoop components like Kafka, Spark, Hive, Sqoop and Zookeeper using Programming languages Python and Scala.
- ▶ worked together with cross-functional teams to collect business needs and use BI dashboards and reports to provide actionable insights.
- ▶ BI solutions were troubleshoot and optimized to guarantee system dependability, high performance, and accurate data.
- ▶ promoted best practices in data modeling, SQL development, and the use of BI tools by mentoring and advising new engineers.
- ▶ Designed, deployed, and configured Microsoft Azure for a multitude of applications utilizing the Azure stack (Including Virtual Machines, Data Factory, Azure Functions, Azure Storage, Azure SQL Database, Azure Event Hubs, Azure Data Lake Storage), focusing on high-availability, fault tolerance, and auto-scaling.
- ▶ Written Terraform scripts to automate Azure services which include Load Balancers, Virtual Networks, Virtual Machines, SQL Databases, Blob Storage, Azure Functions, and converted existing Azure infrastructure to Azure Functions deployed via Terraform.
- ▶ Knowledge of troubleshooting failures in spark applications and fine-tuning for better performance. Particularly valuable experience in complete project life cycle (design, development, testing and implementation).
- ▶ Experience working with different databases such as Oracle, SQL Server, and MySQL, writing stored procedures, functions, joins, and triggers for different Databases

Technical Skills:

Cloud Platforms	GCP (Big Query, Dataflow, Composer, Cloud Storage, Pub/Sub), Azure (ADLS, ADB, Data Factory, Synapse Analytics, Function Apps)
Big Data Technologies	HDFS, Spark (Core, SQL, Data Frames), Hive, Sqoop, Oozie, HBase, Kafka, Palantir, Hortonworks (HDP), Hadoop, AWS (EMR, Glue), GCP (Composer), Azure HDInsight
Data Processing	PySpark, Spark SQL, Hive QL, Python, Scala
Data Warehousing	Snowflake, Redshift
Big Data Platforms	Cloudera, Hortonworks, Palantir
Databases/ RDBMS	MySQL, SQL/PL-SQL, MS-SQL Server 2005, Neo4j, Oracle 9i/10g/11g, Cassandra, MongoDB
Programming Languages	Java, Scala, Python
Software Development	ETL Development (Informatica PowerCenter, SSIS), Airflow
Cloud Development/Deployment	Kubernetes, Docker, Cloud Functions
CI/CD	Jenkins, Code Build, Code Pipeline
Version Control	Git, Bitbucket, Code Cloud
Data Catalog	Google Cloud Platform Data Catalog
Testing	Load Runner, Quick Test Pro
Data Analysis/Visualization	Tableau, Power BI
Software Life Cycles	SDLC (Software Development Life Cycle), Agile models

Professional Experience

Client: UnitedHealth Group, Atlanta,GA

May 2024 - Present

Role: Senior Data Engineer

Project Description: UnitedHealth Group aim is to be designing, building, and maintaining scalable data pipelines on Google Cloud Platform (GCP) to support large-scale data processing and analytics needs. The project involved developing a cloud-based data architecture using GCP services to ingest, transform, and store diverse data types. I collaborated with cross-functional teams to optimize the data ecosystem for real- time and batch processing, providing insights into financial and insurance datasets.

Responsibilities:

- Designed and developed data pipelines using Google Cloud Dataflow, Apache Beam, and Google Cloud Pub/Sub for real-time data streaming and batch processing.
- Built and managed scalable ETL processes utilizing Cloud Composer (Apache Airflow) for workflow orchestration and job scheduling.

- Developed and automated data pipelines using Python, Pandas, and NumPy, improving efficiency in data processing and transformation. Implemented Google Cloud Storage (GCS) to handle raw data ingestion and intermediate storage, ensuring cost-effective and scalable solutions for long-term storage.
- Collaborated with Data Science teams to create machine learning models, ensuring smooth integration of data pipelines with AI Platform.
- Performed data wrangling, cleansing, and transformation using DataPrep, SQL, and Python, to prepare datasets for analytics and reporting.
- Leveraged Cloud Functions and Cloud Run to implement serverless computing solutions for event-driven data processing tasks.
- Integrated Python with cloud-based data platforms like BigQuery, Snowflake, and Azure Synapse for efficient data handling. Automated infrastructure provisioning and management using Terraform for cloud resource orchestration, focusing on cost optimization and operational efficiency.
- Developed interactive dashboards and reports using Power BI, Looker, and Tableau, providing real-time insights for stakeholders.
- Provided ongoing support, troubleshooting, and performance tuning for the GCP data infrastructure to ensure high availability and security of the environment.

Environment: GCP (Google Cloud Platform), BigQuery, Dataflow, Pub/Sub, Cloud Composer, Cloud Storage, Cloud Functions, Kubernetes (GKE), Airflow, Python, SQL, Terraform, Jenkins, Git, Data Studio, Looker

Client: Sollis Health, San Francisco, CA

April 2023 – April 2024

Role: GCP Data Engineer

Project Description: Sollis Health aims to enhance the quality of patient care and operational efficiency by leveraging data engineering techniques to develop an integrated healthcare analytics platform. This platform will integrate patient data, clinical insights, and operational metrics to provide personalized, data-driven healthcare services.

Responsibilities:

- Built scalable distributed data solutions using GCP Big Data Technologies.
- Led the design and development of end-to-end data pipelines using PySpark for ingesting, transforming, and loading large-scale retail data from diverse sources into GCS and BigQuery.
- Mentored junior team members on PySpark best practices and data engineering principles.
- Experienced in loading and transforming large sets of structured and semi-structured data.
- Provided Looker training sessions to business users, empowering them to create and modify self-service dashboards.
- Designed and developed end-to-end ETL solutions and processing applications for streaming and batch data.
- Handled importing of data from various data sources, performed transformations using Spark, and loaded data into GCS.
- Implemented Spark using Scala and SQL for faster testing, processing, and extraction/transformation of data.
- Utilized Apache Spark (SQL & Scala) within Google Cloud Dataproc and GCP environment for data processing, transformation, and pipeline automation.
- Performed unit testing, integration testing, functional testing, usability testing, stress testing, performance testing using automated tools.
- Redesigned views and created stored procedures in BigQuery for performance optimization.
- Defined virtual warehouse sizing in BigQuery for various workloads and unit tested data between Teradata and BigQuery.
- Developed APIs for data source integration and implemented cloud-based integration with Elastic.
- Analyzed large datasets using BigQuery SQL queries and employed various performance optimization techniques.
- Scheduled and automated workflows using Apache Airflow.
- Identified data sources and developed pipelines using PySpark for data normalization into BRDs (Business Ready Datasets).

- Automated data acquisition and BRD creation into a production environment with monitoring.
- Created Python Flask login and dashboard with Neo4j graph database and executed Cypher queries for data analytics.
- Created complex JIRA workflows including project workflows, custom files, notification schemes, reports, and dashboards.
- Used Apache Airflow in GCP Composer environment to build data pipelines with various operators and deployed applications to GCP using Spinnaker.
- Built, maintained, and tested infrastructure to aggregate critical business data into Google Cloud Platform (GCP) BigQuery and GCS.
- Developed REST APIs using Python with Flask/Django frameworks and integrated data sources like RDBMS, JDBC, spreadsheets, and text files.
- Analyzed datasets to perform logical analysis, debug data quality, cleanse and transform data, and create reports.
- Generated reports in Neo4j graph database.
- Scheduled and monitored builds in Data-Lineage and configured email alerts for pipeline failures.
- Contributed to an Agile team of developers focusing on data ingestion across multiple sources.
- Worked in a CI/CD environment and used Code-cloud to store code and Jira Align for tracking features, stories, and tasks.

Environment: GCP, Spark, Scala, Hive, HDFS, Sqoop, Oozie, Snow Pipe, Snow SQL, Snowflake, Teradata, Airflow, Docker, Big Query, GCP Storage, Load Runner, Quick Test Pro, MySQL, Neo4j, Python, Windows Batch Scripts, Flask, Django, Code-cloud, Jira Align, CI/CD environment.

Client: Comerica, Dallas

Sep 2020 – March 2023

Role: Sr Data Engineer

Project Description: Comerica Bank aims to improve customer satisfaction and operational efficiency by leveraging data engineering techniques to analyze customer data and deliver personalized banking services. The objective of this project is to develop a customer-centric data platform that integrates customer data from multiple sources, analyzes customer behavior, and enables targeted marketing and service offerings.

Responsibilities:

- Spearheaded the development of Scala code for data transformations and aggregations on large datasets using Spark libraries, enabling efficient data processing for business intelligence needs in Azure Databricks.
- Optimized existing Spark jobs on Azure HDInsight clusters by implementing partitioning and bucketing techniques, resulting in demonstrably faster processing times.
- Utilized T-SQL expertise to design and query data stored in Azure Synapse Analytics, empowering business analysts with valuable insights.
- Leveraged Apache Spark running on Azure HDInsight clusters for large-scale data processing tasks.
- Developed Scala code to perform data transformations and aggregations on datasets using Spark libraries.
- Optimized Spark jobs for performance by leveraging partitioning, bucketing, and other techniques.
- Designed Hive tables with appropriate schema for efficient data storage and retrieval in Azure Synapse Analytics.
- Developed Azure Functions triggered by data streams in Azure Event Hubs for real-time data processing.
- Designed and implemented data lake pipelines on Azure using services like Azure Data Lake Storage, Azure Synapse Analytics, and Azure Databricks.
- Proficient in using WhyLabs, Splunk, and Datadog for model observability, logging, and alerting.
- Strong understanding of MLOps best practices, including CI/CD for ML, versioning, and automated retraining pipelines.

- Leveraged Azure security best practices to ensure secure data storage, access control, and encryption throughout the data pipeline.
- Migrated data from on-premises sources to Azure Data Lake Storage for centralized storage and efficient access.
- Experienced in integrating machine learning models into **BI tools** like Power BI or Tableau for reporting and visualization.
- Developed **data-driven insights** by embedding ML predictions into dashboards for real-time decision-making. Integrated Azure Event Hubs with downstream processing pipelines for near real-time analytics.
- Implemented Azure DevOps for continuous integration and continuous delivery (CI/CD) of data pipeline code on Azure.
- Utilized Azure HDInsight or Azure Databricks to build scalable data pipelines for data transformation and loading into Azure Synapse Analytics.
- Scheduled and orchestrated data pipelines using Azure Data Factory with Azure DevOps for a familiar workflow.
- Implemented Azure Monitor for monitoring pipeline execution, resource utilization, and performance metrics.
- Performed troubleshooting and optimization of data pipelines on Azure to improve processing times and resource efficiency.
- Used Azure DevOps for storing code, code reviews, and tracking project progress in an Agile methodology. Conducted bi-weekly meetings with the team for tracking project milestones, technical discussions, planning, and estimation. Created project-related design documents and identified technology-related risks and issues.

Environment: Spark, Scala, Hive, Azure Synapse Analytics (formerly SQL Data Warehouse), Azure Databricks, Azure Data Lake Storage, Azure Event Hubs, Azure Functions, Azure Resource Manager (ARM) templates, Azure CLI, Azure DevOps, Azure HDInsight, Azure Data Factory, Azure Monitor, Azure Kubernetes Service (AKS), Azure Cloud Functions, T-SQL.

Client: Auto Club Group, Irving, Texas

July 2018 – August 2020

Role: Data Engineer

Project Description: The Auto Club Group aims to enhance member engagement and retention by leveraging data-driven insights to deliver personalized experiences and services. The objective of this project is to develop a member engagement and retention platform that utilizes data engineering techniques to analyse member data, predict member behavior, and optimize member interactions.

Responsibilities:

- Engaged in data ingestion, pipeline development, and security considerations within a data engineering environment.
- Implemented and supported various data and analytics initiatives in collaboration with the technical lead.
- Designed solutions based on stakeholder requirements gathered through discussions.
- Developed Python and SQL code to collect, parse, and process incoming data from multiple vendors, creating views (in JSON format) for downstream or CRM teams.
- Managed data ingestion into storage systems from diverse sources.
- Created pipelines using resources to ingest data from multiple sources, apply transformations, and store processed data in a data lake or RDBMS for downstream applications.
- Optimized the performance of data processing jobs by incorporating techniques like mapping and partitioning.
- Conducted testing of developed pipelines across different environments (dev, test, prod) to ensure desired outcomes.
- Documented successfully created pipelines for access by the CRM team.
- Managed code versioning and updates, and maintained job details, ensuring stakeholders were informed about completed jobs and resolved errors.

- Proficient in Continuous Integration/Delivery (CI/CD) concepts and tools, with exposure to ETL/ELT.
- Developed integrations using REST and SOAP APIs, JDBC connections, and web services.
- Experienced in information architecture and data modeling.
- Proficient in utilizing data modeling software to design logical and physical data models for complex databases, ensuring data integrity.
- Translated business requirements into actionable data models, resulting in optimized database structures and enhanced data management processes.
- Managed instances for various database engines including MySQL, PostgreSQL, Oracle, SQL Server, and MariaDB.

Environment: Data Engineering, Python, SQL, JSON, RDBMS, Data Lake, CRM, Continuous Integration/Delivery (CI/CD), ETL/ELT, REST, SOAP APIs, JDBC, Information Architecture, Data Modeling, Database Management Systems (MySQL, PostgreSQL, Oracle, SQL Server, MariaDB).

Client: State Of Texas, Tx

May 2015 – June 2018

Role: Hadoop Developer

Project Description: Led the development of a robust and user-friendly Electronic Health Records (EHR) system aimed at optimizing healthcare providers' efficiency in managing patient records. Designed and implemented a comprehensive platform that ensures secure storage, seamless retrieval, and efficient management of patient information. The system prioritizes user experience, streamlining data entry, and retrieval processes, ultimately enhancing the overall workflow of healthcare professionals. Project is industry-leading security measures to safeguard sensitive patient data, ensuring compliance with healthcare regulations.

Responsibilities:

- Responsible for ingesting large volumes of data into Hadoop Data Lake Pipeline.
- Developed Spark programs to Transform and Analyze the data.
- Involved in developing Spark programs to parse the raw data, populate staging tables and store the refined data in partitioned table in the EDW.
- Build data pipeline using Kafka and developed Kafka producers to stream data from external source to Kafka Topics.
- Developed Kafka Consumer APIs in Scala for consuming data from Kafka topics and process consumed data using Spark and load the processed streams to HIVE for analytics.
- Developed Spark applications using Scala for performing data cleansing, event enrichment, data aggregation and data preparation needed for reporting teams to consume.
- Experienced in handling large datasets using Spark in Memory capabilities, using Broadcast variables in Spark, efficient Joins, transformations, and other capabilities.
- Involved in creating HIVE tables, imported processed data into tables and done analytics as per business developments.
- Read data from various sources like Oracle and Hive using Apache Spark and process consumed data using Spark and load the process to HIVE and downstream applications for different purposes.
- Developed Apache spark application with Python to perform distinct kinds of validations and standardization on fields based on certain validations rules on incoming Data.
- Developed required UDFs using python scripts and implemented them in application.
- Used Spark RDD's, Data Frames and Datasets Spark- SQL faster processing of ETL workloads.
- For performance optimization I implemented different optimization techniques like Partition, Bucketing and Map Side joins for fast query processing.
- Implemented Spark with Python and Spark SQL for faster testing and processing of data.
- Generated Python Scripts for retrieval and analysis on No-SQL database such as HBase.
- Used NOSQL HBase for storing batching information.
- Used Rally tracking tool for assigning and defect management.

- Responsible of web application deployments over cloud services (web and worker roles) on Azure, using VS and PowerShell
- Worked in Agile development environment and actively involved in daily Scrum and other design related meetings. Used Bitbucket to share the code snippet among the team members.
- Provided high availability for IaaS VMs and PaaS role instances for access from other services in the VNet with Azure Internal Load Balancer.
- Implemented high availability with Azure Classic and Azure Resource Manager deployment models and worked on creating Custom Azure Templates for quick deployments and advanced PowerShell scripting.
- Migrated SQL Server 2008 database to Windows Azure SQL Database and updated the Connection Strings.

Environment: Spark, Oracle, HBase, Python, Bitbucket, Rally, Azure, SQL Server, Hive, Kafka, Scala, SQL, Spark SQL, Visual Studio, PowerShell, Agile methodologies, Scrum, Azure Internal Load Balancer, Windows Azure SQL Database.

Client: Ameriprise, Minneapolis

Jan 2013 – April 2015

Role: SQL Developer

Project Description: The Automated Investment Advisory platform successfully democratized access to personalized financial advice, empowering users to make informed investment decisions. This project involved creating a user-centric and technologically advanced solution to provide automated investment advice tailored to individual user preferences and financial goals. Its user-friendly interface, backed by robust backend processes and personalized algorithms, garnered positive feedback and increased user engagement.

Responsibilities:

- Designed ETL Process using Informatica to load data from Flat Files, and Excel Files to target Oracle Data Warehouse database.
- Interacted with the business community and database administrators to identify the Business requirements and data realities.
- Experienced in Agile Methodologies and SCRUM Process. Formulated and executed test queries using SQL on Oracle and SQL Server Databases.
- Built various graphs for business decision making using Python matplotlib library.
- Worked in development of applications especially in UNIX environment and familiar with all its commands.
- Implement code in Python to retrieve and manipulate data. Reviewed basic SQL queries and edited inner, left, & right joins in Tableau Desktop by connecting live/dynamic and static datasets.
- Reported and created dashboards for Global Services & Technical Services using SSRS, Oracle BI, and Excel. Deployed Excel VLOOKUP, PivotTable, and Access Query functionalities to research data issues.
- Created conceptual, logical, and physical relational models for integration and base layer; created logical and physical dimensional models for presentation layer and dim layer for a dimensional data warehouse in Power Designer.
- Involved in reviewing business requirements and analyzing data sources from Excel, Oracle SQL Server for design, development, testing, and production rollover of reporting and analysis projects.
- Analyzing, designing, developing, implementing, and maintaining ETL jobs using IBM Info sphere Data stage and Netezza.
- Designed Data Flow Diagrams, E/R Diagrams and enforced all referential integrity constraints.
- Extensively worked in Client-Server application development using Oracle 10g, Teradata 14, SQL, PL/SQL, Oracle Import and Export Utilities.
- Coordinated with DB2 on database build and table normalizations and de-normalizations.
- Conducted brainstorming sessions with application developers and DBAs to discuss various de-normalization, partitioning and indexing schemes for Physical Model.

- Involved in several facets of MDM implementations including Data Profiling, metadata acquisition and data migration.
- Extensively used SQL Loader to load data from the Legacy systems into Oracle databases using control files and used Oracle External Tables feature to read the data from flat files into Oracle staging tables.
- Involved in extensive Data validation by writing several complex SQL queries and involved in back-end testing and worked with data quality issues.
- Used SSIS to create ETL packages to validate, extract, transform and load data to data warehouse databases, data mart databases, and process SSAS cubes to store data to OLAP databases.
- Created ETL packages using OLTP data sources (SQL Server, Flat files, Excel source files, Oracle) and loaded the data into target tables by performing various kinds of transformations using SSIS.
- Migrated SQL server 2008 to SQL Server 2008 R2 in Microsoft Windows Server 2008 R2 Enterprise Edition.
- Developed and published reports and dashboards using Power BI and written effective DAX formulas and expressions. Utilized Power Query in Power BI to pivot and un-pivot the data model for data cleansing and data massaging.
- Developing reusable objects like PL/SQL program units and libraries, database procedures and functions, database triggers to be used by the team and satisfying the business rules.
- Performed data validation on the flat files that were generated in UNIX environment using UNIX commands, as necessary.
- Strong understanding of Data Modeling (Relational, dimensional, Star and Snowflake Schema), Data analysis, implementations of Data warehousing using Windows and UNIX
- Used Model Mart of Erwin for effective model management of sharing, dividing, and reusing model information and design for productivity improvement and identified and tracked the slowly changing dimensions and determined the hierarchies in dimensions.
- Designed elevated level ETL architecture for overall data transfer using SSIS from the source server to the Warehouse and defined various facts and dimensions in the data mart including Fact less Facts and designed the Data Mart defining Entities, Attributes, and relationships between them.

Environment: MySQL, HBase, Jira,ETL, PL/SQL, SSIS, OLAP, UNIX, DAX.

Educational Summary:

- Bachelor's in computer science in Acharya Nagarjuna University - 2012 (INDIA).